WORKED SOLUTION

STEP II 1998, Question 14

Probability and Statistics • Binomial and Poisson Expectation

STEP II 1998, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q14

The key idea

Despite the database label, this is a probability question: Dr Doom should take the new scheme if her *expected* salary under it beats the fixed salary A. The number of failures is binomial, so we need $\mathrm{E}[1/(1+X)]$. The awkward factor $\frac{1}{k+1}$ is absorbed by the identity $\frac{1}{k+1}\binom{N}{k} = \frac{1}{N+1}\binom{N+1}{k+1}$, which turns the sum into an almost complete binomial expansion of $(p+q)^{N+1}$. The same trick, with $\frac{\lambda^k}{(k+1)!}$, handles the Poisson model.

Solution

When to accept the new scheme

Let X be the number of Dr Doom's students who fail, and write $q = 1-p$. The students fail independently, each with probability p, so $X \sim \mathrm{B}(N,p)$:

$$\mathrm{P}(X=k) = \binom{N}{k} p^k q^{N-k}, \qquad k = 0,1,\ldots,N.$$

We assume Dr Doom is content to judge the schemes by expected salary. Under the new scheme her salary is the random variable $S = B/(1+X)$, so

$$\mathrm{E}(S) = B\sum_{k=0}^{N} \frac{1}{k+1}\binom{N}{k} p^k q^{N-k}.$$

Ask yourself How do we get rid of the $\frac{1}{k+1}$ inside the sum?

Use factorials:

$$\begin{aligned}\frac{1}{k+1}\binom{N}{k} &= \frac{N!}{(k+1)!\,(N-k)!} = \frac{1}{N+1}\cdot\frac{(N+1)!}{(k+1)!\,((N+1)-(k+1))!} \\ &= \frac{1}{N+1}\binom{N+1}{k+1}.\end{aligned}$$

Substituting this, and multiplying and dividing by p so that the power of p matches the new binomial coefficient,

$$\begin{aligned}\mathrm{E}(S) &= \frac{B}{(N+1)p}\sum_{k=0}^{N}\binom{N+1}{k+1}p^{k+1}q^{N-k}\\ &= \frac{B}{(N+1)p}\sum_{j=1}^{N+1}\binom{N+1}{j}p^{j}q^{N+1-j}\\ &= \frac{B}{(N+1)p}\left[(p+q)^{N+1}-q^{N+1}\right]\end{aligned}$$

putting $j = k+1$ and then using the binomial theorem, since the sum is the full expansion of $(p+q)^{N+1}$ with only the $j=0$ term, q^{N+1}, missing. As $p+q=1$,

$$\mathrm{E}(S) = \frac{B(1-(1-p)^{N+1})}{(N+1)p}.$$

She should accept if $\mathrm{E}(S) > A$. Multiplying through by $(N+1)p$, which is positive (we need $p>0$ for the formula to make sense; if $p=0$ she simply compares B with A), this is

$$A(N+1)p < B(1-(1-p)^{N+1}),$$

which is the required condition.

The Poisson model

A binomial distribution $\mathrm{B}(N,p)$ is well approximated by a Poisson distribution with mean $\lambda = Np$ when N is large and p is small, with Np of moderate size. So Dr Doom's failures could be modelled as Poisson if she has a large class, each student is unlikely to fail, and failures still occur independently. (More generally, a Poisson model suits failures that occur singly, independently and at a constant average rate.)

With $X \sim \mathrm{Po}(\lambda)$, $\lambda = Np$, the same idea works, now with $\frac{1}{k+1}\cdot\frac{1}{k!} = \frac{1}{(k+1)!}$:

$$\begin{aligned}\mathrm{E}\Big(\frac{B}{1+X}\Big) &= B\sum_{k=0}^{\infty}\frac{1}{k+1}\cdot\frac{\mathrm{e}^{-\lambda}\lambda^{k}}{k!} = \frac{B\mathrm{e}^{-\lambda}}{\lambda}\sum_{k=0}^{\infty}\frac{\lambda^{k+1}}{(k+1)!}\\ &= \frac{B\mathrm{e}^{-\lambda}}{\lambda}(\mathrm{e}^{\lambda}-1) = \frac{B(1-\mathrm{e}^{-\lambda})}{\lambda},\end{aligned}$$

since the last sum is the exponential series with its first term, 1, missing. So Dr Doom's expected salary under this model is

$$\frac{B(1-\mathrm{e}^{-Np})}{Np} \text{ pounds.}$$

Checks

- $N=1$: directly, $\mathrm{E}(S) = B\big(q+\frac{1}{2}p\big) = B\big(1-\frac{1}{2}p\big)$, and the formula gives $B\,\frac{1-q^2}{2p} = B\,\frac{1+q}{2} = B\big(1-\frac{1}{2}p\big)$.

- $p \to 0$: the binomial expression tends to B (no one fails), and so does the Poisson one as $\lambda \to 0$.
- The two answers agree when the approximation should be good: with $N = 1000$, $p = 0.003$, $\mathrm{E}(S)/B$ is 0.31655 from the binomial formula and 0.31674 from the Poisson one. Exact enumeration and a simulation with $N = 20$, $p = 0.3$ both give $\mathrm{E}(S)/B \approx 0.1586$.

Common slips

- Writing $\mathrm{E}\left(\frac{1}{1+X}\right) = \frac{1}{1+\mathrm{E}(X)} = \frac{1}{1+Np}$. Expectation does not pass through non-linear functions; in fact $\frac{1}{1+x}$ is convex, so by Jensen's inequality the true value is larger.
- Forgetting that the re-indexed sum misses the $j = 0$ term, giving $\mathrm{E}(S) = B/((N+1)p)$.

Beyond STEP

The trick here is a special case of computing $\mathrm{E}\big(1/(1+X)\big)$ via $\frac{1}{1+x} = \int_0^1 t^x \,\mathrm{d}t$, which links it to the probability generating function $G(t) = \mathrm{E}(t^X)$: the answer is $\int_0^1 G(t)\,\mathrm{d}t$. Try it with $G(t) = (q+pt)^N$ and $G(t) = \mathrm{e}^{\lambda(t-1)}$. Generating functions, and the convergence of binomial to Poisson, are covered well in Grimmett and Welsh, *Probability: An Introduction*.

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